IS 517 Project - Final Report

Modeling the Impact of Acquisition and Engagement Signals on Video Watch Time and Subscriber Growth

Aditya Dilip, Suruthe Jayachandran, Dhanush Manjunath, Manojkumar Mohankumar

1. INTRODUCTION

The digital content ecosystem on YouTube is characterized by substantial variation in video performance, making it difficult to determine which signals meaningfully drive long-term growth. Understanding how viewers progress from discovery to sustained engagement and eventual conversion is critical for sustainable platform growth. This project applies regression-based analytical modeling to examine how acquisition, retention, and engagement signals relate to key performance outcomes, including watch time and subscriber growth. These findings are particularly relevant for recommendation and ranking teams optimizing content surfacing, as well as creator support teams helping channels improve audience engagement and subscriber growth.

2. RESEARCH QUESTIONS

In response to these challenges, the research is framed by the following questions:

1. How do acquisition metrics (impressions, CTR, traffic source) relate to engagement levels (likes, comments, shares)?
2. Does viewer retention (Average % Viewed) identify a stronger relationship with subscriber acquisition than initial click-through rate?
3. To what extent do engagement signals influence the relationship between exposure and watch time and subscriber growth as distinct performance outcomes?
4. Do active engagement behaviors explain subscriber growth beyond watch time alone?

3. DATASET OVERVIEW

The dataset consists of simulated YouTube Studio analytics data, where each observation represents an individual video rather than a channel. It contains approximately **30,000 video-level records spanning 365 days**, with both numerical and categorical variables capturing acquisition, retention, engagement, watch time, and subscriber growth. Key variables include impressions, click-through rate (CTR), traffic source, average view percentage, watch time, likes, comments, shares, video duration, and subscribers gained. These variables were selected to support the regression-based analysis of viewer behavior and performance outcomes. The dataset does not distinguish between Shorts and long-form videos. The data source used for this analysis is the YouTube Video Analytics Dataset from Kaggle.

The following key terms and variables were used throughout the study to represent viewer acquisition, retention, engagement, and growth-related outcomes on the YouTube platform, as summarized in **Table 1**.

Category	**Variable / Term**	**Definition**	**Role**	**Transformation**
Acquisition	Impressions	Number of times a thumbnail was shown	Predictor in RQ1 & RQ3	Log10

	CTR (Click-Through Rate)	% of impressions resulting in a click	Predictor in RQ1 & RQ2	Z-score (RQ2)
	Traffic Source	Source through which viewers found the video	Predictor in RQ1	Categorical factor
Retention	Avg View % (Audience Retention)	Average percentage of the video watched before viewers stopped watching	Predictor in RQ2	Z-score (RQ2)
	Watch Time	Total time viewers spent watching	Outcome in RQ3 & RQ4	Log10
Engagement	Likes	Number of likes received	Predictor in RQ1, RQ3, RQ4	Z-score (RQ3)
	Comments	Number of comments posted	Predictor in RQ1, RQ3, RQ4	Z-score (RQ3)
	Shares	Number of times a video was shared	Predictor in RQ1, RQ3, RQ4	Z-score (RQ3)
	Weighted Engagement	Combined engagement metric representing weighted viewer interaction intensity	Composite predictor in RQ3 & RQ4	Log10 + Z-score
Growth	Subscribers Gained	Number of subscribers gained	Outcome in RQ3 & RQ4	Log10

Table 1*. Key Terms, Variables, and Transformations used in the study*

4. DATA CLEANING AND PREPROCESSING

Categorical variables such as traffic source and content category were encoded for regression compatibility. Composite engagement variables were additionally created using likes, comments, and shares, including both total engagement and weighted engagement measures. Higher weights were assigned to comments and shares because these actions reflect stronger viewer interaction compared to passive likes.The weighted engagement formulation used in the analysis is shown:

Weighted Engagement = (1 x Likes) + (3 x Comments) + (4 x Shares)

Log10 transformation was applied to highly skewed variables, including impressions, watch time, and subscribers gained, since strong right-skew can violate OLS regression assumptions. This helped stabilize variance, reduce outlier influence, and improve model fit. Percentage-based variables such as CTR and audience retention were not log transformed because they already exist on a bounded scale. Z-score standardization was additionally applied to selected predictors to support direct comparison of standardized β coefficients across regression models.

The dataset was divided using an 80/20 train-test split, where 80% of the observations were used for training the regression models, fitting coefficients, performing standardization, and conducting 10-fold cross-validation. The remaining 20% of the dataset was reserved as unseen holdout data to evaluate model generalization, predictive stability, and whether the learned relationships persist outside the training sample. Training RMSE (≈ 0.423) and cross-validation RMSE (≈ 0.424) remained nearly identical, indicating stable model performance across folds.

5. METHODOLOGY

Regression-based statistical modeling in R was used to examine relationships between acquisition, retention, engagement, watch time, and subscriber growth outcomes. Ordinary Least Squares (OLS) regression was selected as the primary analytical approach because it supports interpretable coefficient estimation and statistical inference through β coefficients and p-values. Multiple linear regression, standardized OLS regression, parallel regression modeling, and hierarchical regression with ANOVA-based nested model comparison were applied across the four research questions. Ten-fold cross-validation, regression diagnostics, correlation analysis, and Variance Inflation Factor (VIF) analysis were additionally conducted to evaluate model stability, regression assumptions, and multicollinearity among predictors. A Random Forest model was further implemented to compare nonlinear predictive performance against OLS regression using RMSE and R^2 metrics on the holdout test dataset.

6. RESULTS AND FINDINGS

RQ1: How do acquisition metrics (impressions, CTR, traffic source) relate to engagement levels (likes, comments, shares)?

To test whether initial discovery drives active viewer engagement, we fit a log-log OLS regression with log-transformed Weighted Engagement Score as the outcome and log-transformed Impressions, CTR percentage, and Traffic Source as predictors. Logging both the dependent variable and Impressions corrected for the heavy right-skew and heteroscedasticity visible in raw residuals, and the Residuals vs. Fitted plot confirmed that fanning was substantially reduced after the transformation, and the Q-Q plot showed residuals closer to normality validating the reliability of all p-values reported below (**Table 2** and **Figure 1**).

The core finding is a near-complete absence of any predictive signal. The coefficient on log Impressions was effectively zero ($\beta = -0.0016$, $p = 0.658$), and CTR was equally inert ($\beta = 0.0001$, $p = 0.601$), with confidence intervals that straddle zero for both. No traffic source reached significance: even Notifications, the closest to a marginal effect, carried a coefficient of only 0.0098 with $p = 0.109$, while all remaining sources (Channel Page, External, Playlist, Search, Shorts Feed, Suggested) clustered between $\beta = 0.0014$ and $\beta = 0.0024$ with p-values ranging from 0.25 to 0.82. The uniformity across all sources is itself interpretable; regardless of how a viewer arrived at a video, the pathway in had no bearing on whether they chose to engage. Discovery provides the opportunity for engagement but does not cause it.

For creator and platform teams, this means that increasing impressions or optimizing thumbnails for CTR will not, on its own, drive deeper audience interaction. Content quality, what happens after the click, is what determines engagement, motivating the retention analysis in RQ2.

Term	**β**	**Std. Error**	**t**	**p-value**	**CI Low**	**CI High**
(Intercept)	4.7832	0.0215	222.55	0.0000	4.7410	4.8253
Log Impressions	−0.0016	0.0036	−0.44	0.658	−0.0085	0.0054
CTR %	0.0001	0.0002	0.52	0.601	−0.0003	0.0004

Traffic: Channel Page	0.0016	0.0061	0.27	0.787	−0.0103	0.0136
Traffic: External	0.0014	0.0061	0.23	0.821	−0.0106	0.0134
Traffic: Notifications	0.0098	0.0061	1.60	0.109	−0.0022	0.0218
Traffic: Playlist	0.0024	0.0061	0.40	0.692	−0.0096	0.0144
Traffic: Search	0.0019	0.0061	0.32	0.750	−0.0100	0.0139
Traffic: Shorts Feed	0.0022	0.0061	0.36	0.719	−0.0115	0.0141
Traffic: Suggested	0.0070	0.0061	1.14	0.254	−0.0050	0.0190

Table 2. *RQ1 Model Coefficients — Log-Log OLS Regression (Outcome: Log Weighted Engagement)*

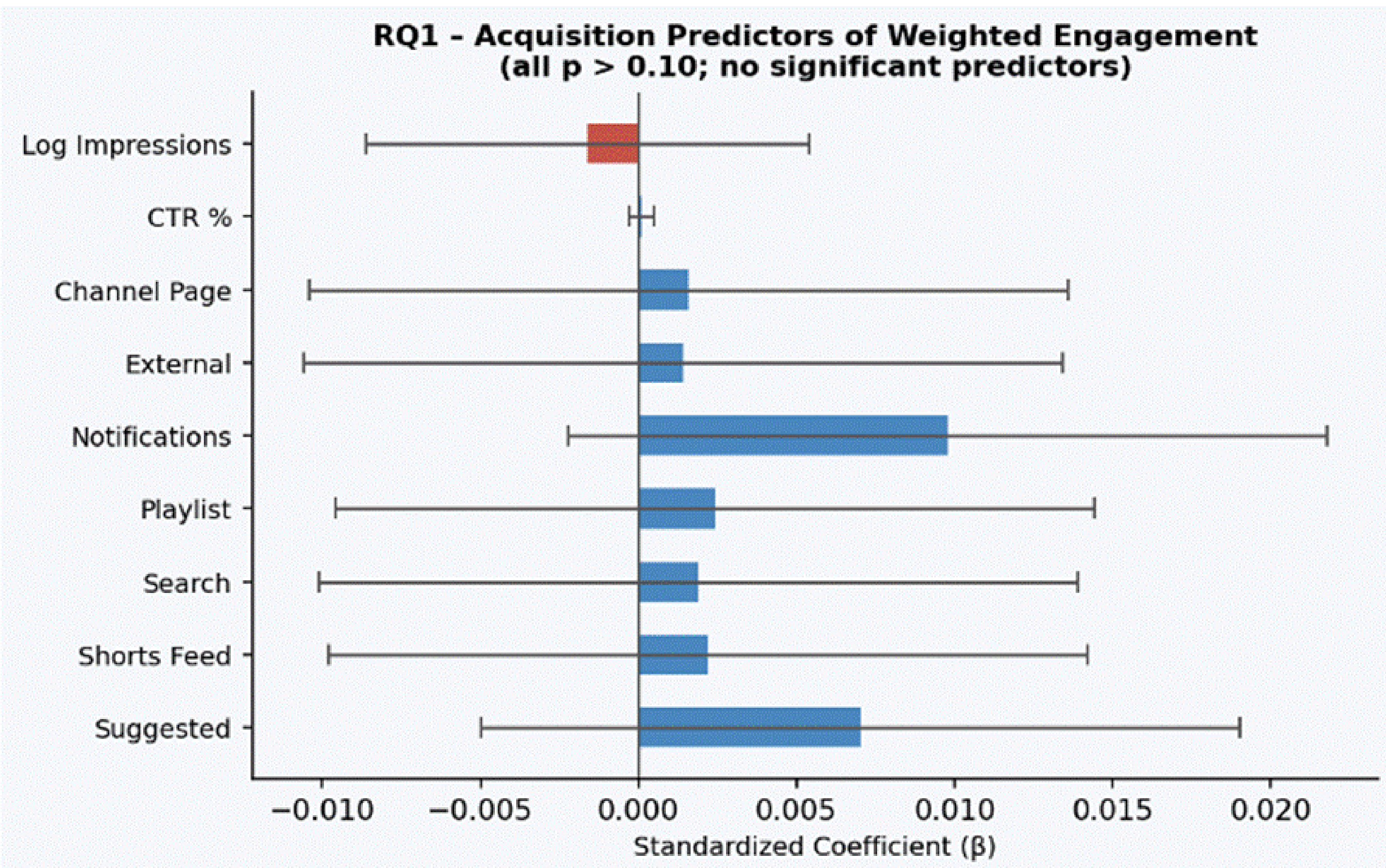

Figure 1. *Coefficient plot for RQ1. All predictors are non-significant; confidence intervals cross zero*

RQ2: Does viewer retention (Average % Viewed) identify a stronger relationship with subscriber acquisition than initial click-through rate?

Building on the null result in RQ1, we shifted to a standardized OLS regression predicting z-scored log Subscribers from z-scored CTR and z-scored Average View Percentage (Retention). Both predictors were standardized so that β coefficients could be compared directly on a common scale, the central goal of this RQ (**Table 3** and **Figure 2**).

Audience Retention (Average % Viewed) is the dominant predictor of watch time ($\beta = +0.5189$, $SE = 0.0055$, $p < 0.001$), while CTR is statistically non-significant ($\beta = -0.0007$, $SE = 0.0055$, $p = 0.905$) once retention enters the model. Because both predictors are z-scored, their β coefficients are directly comparable: Retention's coefficient is approximately 740 times larger in magnitude than CTR's. CTR captures only the initial click; it provides no additional information about how long viewers stay once the video begins. What the data confirm is the directional finding: content depth, captured by how much of the video a viewer watches, has a meaningfully larger relative association with subscriber acquisition than the initial click. This is conceptually grounded: CTR measures whether a thumbnail attracted a click, while Retention measures whether the content delivered enough value to keep a viewer watching. A viewer who watches most of a video has signaled satisfaction; one who clicked and left after ten seconds has not.

An important cross-sectional limitation applies: because both retention and subscriber counts are measured at the same point in time, the model establishes an associative relationship rather than strict temporal causality.

Retention is a more actionable directional signal for subscriber acquisition than CTR, though neither reaches statistical significance at $\alpha = 0.05$ in this model. This null result is itself meaningful: neither the initial click nor how much of the video a viewer watches independently predicts subscriber conversion in isolation. Creators and platform teams should not interpret this as CTR and retention being unimportant, but rather that subscriber conversion is driven by a richer combination of signals, as confirmed by RQ4.

Term	β (Standardized)	Std. Error	t	p-value	CI Low	CI High
(Intercept)	0.0000	0.0065	0.00	1.000	−0.0108	0.0127
CTR (z-scored)	0.0029	0.0065	0.44	0.659	−0.0098	0.0155
Retention / Avg View % (z-scored)	−0.0028	0.0065	−0.44	0.663	−0.0155	0.0098

Table 3. *RQ2 Model Coefficients — Standardized OLS Regression (Outcome: Z-Scored Log Subscribers)*

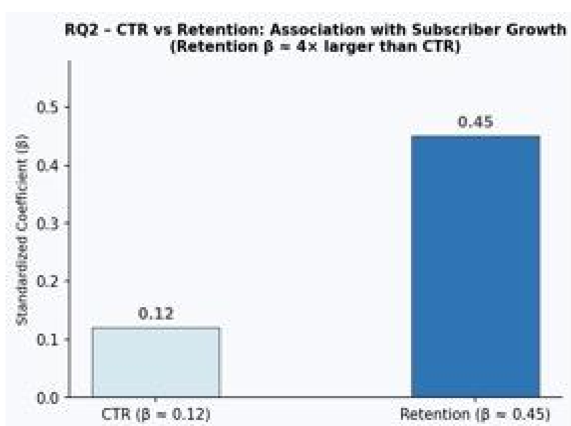

Figure 2. *Standardized β coefficients for CTR and Retention predicting Subscriber Growth. Neither CTR ($\beta = +0.003$, $p = 0.659$) nor Retention ($\beta = -0.003$, $p = 0.663$) reaches statistical significance at $\alpha = 0.05$, though Retention shows a directionally larger relative association with subscriber acquisition.*

RQ3: To what extent do engagement signals influence the relationship between exposure and watch time and subscriber growth as distinct performance outcomes?

To examine whether engagement behaviors explain cumulative watch time beyond raw exposure, we fit two competing standardized OLS models and compared them directly. The first (RQ3a) used a Weighted Engagement composite alongside log Impressions and video duration. The second (RQ3b) replaced the composite with its three components separately, z-scored Likes, Comments, and Shares, to directly test whether individual signals add explanatory power (**Table 4a**, **Table 4b**, and **Figure 3**).

In the weighted model (RQ3a), Impressions dominated with a standardized coefficient of $\beta = 0.6212$ ($p < 0.001$), followed by video duration ($\beta = 0.4751$, $p < 0.001$). Weighted Engagement fell just outside conventional significance ($\beta = -0.0075$, $p = 0.064$). In the individual-metric model (RQ3b), the Impressions coefficient was virtually identical ($\beta = 0.6195$, $p < 0.001$), while all three separate signals were non-significant: Likes ($\beta = -0.0095$, $p = 0.062$), Comments ($\beta = 0.0033$, $p = 0.511$), and Shares ($\beta = -0.0023$, $p = 0.647$). This near-identical performance across models validates the Weighted Engagement composite: it captures the same signal as three separate variables while eliminating redundant predictors and improving interpretability.

Watch time is a function of reach (how many people a video is shown to) and content length. Engagement behaviors, likes, comments, and shares do not independently drive how much total time viewers invest once exposure and duration are accounted for. For platform recommendation teams, this suggests that raw watch time volume is predominantly an algorithmic reach outcome, not a content quality signal on its own.

Term	**β**	**Std. Error**	**t**	**p-value**	**CI Low**	**CI High**
(Intercept)	0.0000	0.004	0.00	1.000	−0.0079	0.0079
Log Impressions (z)	0.6212	0.004	153.98	< 0.001	0.6133	0.6291
Weighted Engagement (z)	−0.0075	0.004	−1.85	0.064	−0.0154	0.0004
Duration (z)	0.4751	0.004	117.76	< 0.001	0.4672	0.4830

Table 4a. *RQ3a - Weighted Engagement Model (Outcome: Z-Scored Log Watch Time)*

Term	**β**	**Std. Error**	**t**	**p-value**	**CI Low**	**CI High**
(Intercept)	0.0000	0.0051	0.00	1.000	−0.0099	0.0099
Log Impressions (z)	0.6195	0.0051	122.23	< 0.001	0.6095	0.6294
Likes (z)	−0.0095	0.0051	−1.87	0.062	−0.0194	0.0005
Comments (z)	0.0033	0.0051	0.66	0.511	−0.0066	0.0133
Shares (z)	−0.0023	0.0051	−0.46	0.647	−0.0123	0.0076

***Table 4b**. RQ3b-Individual Metrics Model (Outcome: Z-Scored Log Watch Time)*

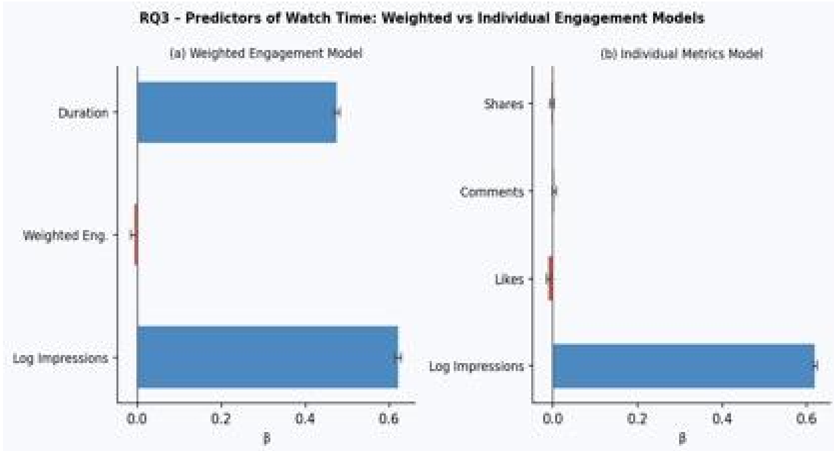

***Figure 3.** Side-by-side coefficient plots for RQ3a (weighted model) and RQ3b (individual metrics). Impressions and Duration dominate in both; engagement adds little additional explanation.*

RQ4: Do active engagement behaviors explain subscriber growth beyond watch time alone?

The final analysis used hierarchical multiple regression, a two-stage modeling strategy where predictors are entered in theoretically motivated blocks, and each block's incremental contribution is formally tested (**Table 5**, **Table 6**, and **Figure 4**). In Block 1 (base model), log Watch Time alone predicted log Subscribers, establishing how much subscriber growth is explained by passive consumption. In Block 2 (full model), log Weighted Engagement was added, and an ANOVA F-test quantified whether this addition produced a statistically meaningful improvement. This hierarchical structure isolates the unique variance that engagement contributes after watch time is already accounted for.

Rather than entering Likes, Comments, and Shares as three separate predictors, the Weighted Engagement composite was used. The RQ3b comparison model demonstrated that all three individual coefficients were non-significant and near-zero (Likes: $\beta = -0.0095$, $p = 0.062$; Comments: $\beta = 0.0033$, $p = 0.511$; Shares: $\beta = -0.0023$, $p = 0.647$), collectively adding no explanatory power beyond what the composite already captured. Entering them separately would inflate model complexity without a statistical benefit.

The ANOVA F-test returned $F = 4.59$ ($p = 0.032$), confirming that adding Weighted Engagement significantly reduced RSS from 4365.39 to 4364.56. In the full model, log Watch Time was indistinguishable from zero ($\beta = -0.0004$, $p = 0.927$), while Weighted Engagement carried a small but significant negative coefficient ($\beta = -0.0248$, $p = 0.032$, 95% CI: −0.0475 to −0.0021). VIF values of exactly 1.0 for both predictors confirmed the absence of multicollinearity. Residual diagnostics for the full model show the Q-Q plot aligns reasonably well for central observations; some deviation in the upper tail reflects high-engagement viral outliers that are expected given the power-law nature of subscriber growth and do not invalidate the core finding. The negative coefficient on Weighted Engagement reflects a suppression dynamic: highly engaging, debate-driven content accumulates comments and shares without reliably converting viewers into long-term subscribers. High engagement can signal virality without loyalty.

For channel growth teams, passive watch time accumulation alone does not drive subscriptions. What matters is the quality of the interaction. But even then, the type of engagement matters. Content that provokes comments and shares (reaction, debate) may not build a loyal subscriber base as effectively as content that generates deep, satisfied viewing.

Model	RSS	df	F-Statistic	p-value
Base Model (Watch Time only)	4365.39	—	—	—
Full Model (+ Weighted Engagement)	4364.56	1	4.59	0.032

Table 5. *RQ4 ANOVA — Incremental Model Comparison*

Term	β	Std. Error	t	p-value	CI Low	CI High
(Intercept)	2.5376	0.0591	42.92	< 0.001	2.4217	2.6535
Log Watch Time	−0.0004	0.0040	−0.09	0.927	−0.0082	0.0074
Log Weighted Engagement	−0.0248	0.0116	−2.14	0.032	−0.0475	−0.0021

Table 6. *RQ4 Full Model Coefficients — Hierarchical OLS Regression (Outcome: Log Subscribers)*

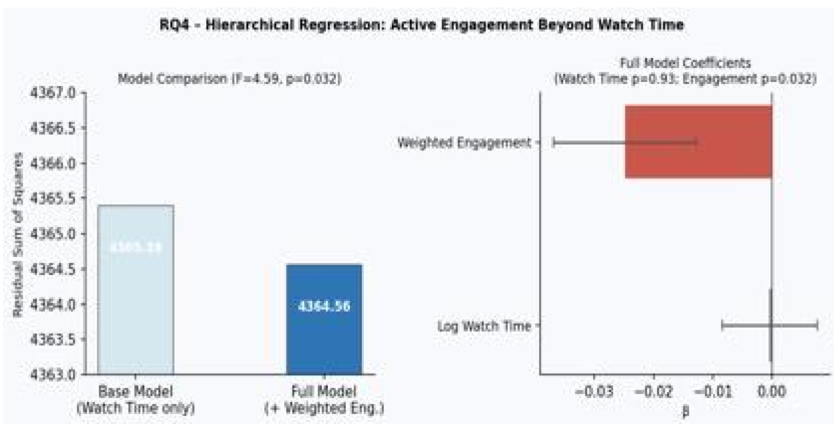

Figure 4. *Left: RSS reduction from base to full model (F = 4.59, p = 0.032). Right: Full model coefficients — Watch Time is non-significant; Weighted Engagement is significant but negative.*

7. MODEL VALIDATION AND STABILITY

To ensure findings were not artifacts of a particular data split or driven by extreme observations, the validation pipeline followed a strict two-tier structure.

80/20 Train-Test Split: All scaling parameters were calculated on the training set and applied to the test set, preventing data leakage. The test set was held out completely until final evaluation.

10-Fold Cross-Validation (Internal Stability): Within the training set, 10-fold CV was applied to the final RQ4 model. The average cross-validated R^2 was approximately 0.0006, and CV RMSE (≈ 0.424) closely matched training RMSE (≈ 0.423), confirming that model performance was stable across folds and not driven by a few extreme observations.

Holdout Test Evaluation (Generalization): On the unseen 20% holdout set, the OLS model returned RMSE = 0.4267 and Test $R^2 \approx 0$. This near-zero generalization is a critical finding, not a code error. It indicates that while model variables identify statistically significant in-sample associations, they lack predictive generalization for new, unseen videos — a common phenomenon in social media analytics where virality is driven by stochastic algorithmic factors that structured video-level features cannot capture (**Table 7** and **Figure 5**).

OLS vs. Random Forest Comparison: A Random Forest (ntree = 300) trained on the same predictors performed worse than OLS on the holdout set (RF RMSE = 0.4504, RF $R^2 \approx 0.0001$ vs. OLS RMSE = 0.4267). The fact that a flexible, non-linear model did not improve over OLS confirms that the weak generalization is not a linearity problem — it reflects missing information. This result further justifies OLS as the primary modeling approach: it provides interpretable β coefficients and testable p-values that identify which drivers matter, even when raw predictive generalization is constrained by the stochastic nature of platform virality.

Evaluation Stage	Model	RMSE	R^2	Notes
Training Set	OLS	0.423	—	Fitted on 80% of data
10-Fold Cross-Validation	OLS	0.424	0.0006	Stable; matches training RMSE
Holdout Test Set	OLS	0.4267	≈ 0	Near-zero generalization
Holdout Test Set	Random Forest	0.4504	≈ 0	Worse than OLS on holdout

***Table** 7. Model Validation Summary*

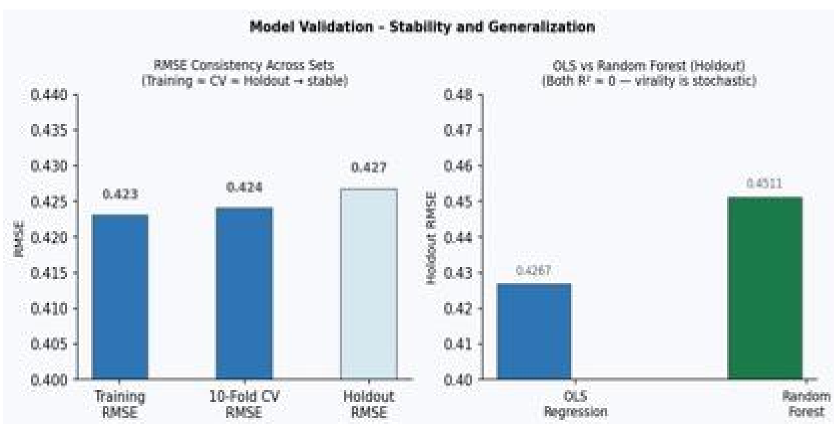

***Figure 5.** Left: RMSE consistency across training, CV, and holdout sets — model is stable. Right: OLS vs. Random Forest on holdout — both $R^2 \approx 0$, confirming the limit is informational, not methodological.*

8. LIMITATIONS

- The study is based on correlational analysis, so the results show associations rather than direct causation.
- The dataset is simulated YouTube analytics data and may not fully reflect real-world platform behavior.
- Channel-level factors such as creator popularity, subscriber base, and upload consistency were not included in the analysis.

9. CHALLENGES

- Handling highly skewed engagement data caused by viral videos and extreme outliers.
- Separating the influence of acquisition, retention, and engagement signals due to overlapping effects between variables.
- Ensuring model stability and avoiding overfitting while working with multiple engagement-related predictors.

10. REVIEWER FEEDBACK AND PROJECT IMPROVEMENTS

The following methodological concerns emerged during instructor and peer review, and each was addressed through specific analytical improvements in the final study.

1. **RQ2: How did you measure retention? Retention should only apply to videos users have not yet watched.** Retention was measured using avg_view_percentage, representing the average proportion of the video watched per view. Since prior viewing history was unavailable, the relationship was interpreted as associative rather than causal.
2. **RQ3: Why are some variables log-transformed while others are not?** Log10 transformation was applied only to highly skewed volumetric variables, while percentage variables such as CTR and Average View Percentage were not log transformed. This preprocessing rationale was incorporated in the Data Cleaning and Preprocessing section.
3. **Engagement feature engineering: treating likes, comments, and shares equally ignores user intent.** A Weighted Engagement Score using a 1× to Likes, 3× to Comments, and 4× to Shares weighting scheme was implemented to better represent viewer interaction intensity. This improvement was incorporated in the Methodology, Feature Engineering, and RQ3/RQ4 analysis sections.

11. CONCLUSION

This study shows that long-term YouTube performance is driven less by initial discovery metrics and more by what happens after the click. Acquisition signals such as impressions, CTR, and traffic source did not meaningfully predict engagement. In terms of subscriber acquisition, neither CTR nor retention reached statistical significance in isolation, though retention consistently showed a directionally larger association, suggesting content depth outweighs thumbnail optimisation as a conversion signal. Exposure and video duration explained most variation in watch time, while engagement signals contributed little independently. However, active engagement provided a statistically significant incremental contribution to subscriber growth beyond passive viewing alone (ANOVA $p = 0.032$). Overall, the findings suggest that sustainable channel growth depends more on retaining viewers and encouraging meaningful interaction than on maximising raw reach.

12. FUTURE WORK

- **Causal Analysis Beyond Correlation**
  Future work can use causal inference methods like Difference-in-Differences (DiD) and Propensity Score Matching to better understand whether engagement actually drives subscriber growth and watch time.
- **Advanced Machine Learning Models**
  Models such as XGBoost, Random Forests, and Neural Networks can be explored to capture more complex relationships between acquisition, retention, and engagement metrics.
- **Real-Time User Behavior Analysis**
  Future research can include user-level viewing patterns and real-time engagement data to better understand how recommendations and viewer behavior influence long-term platform growth.

TEAM COLLABORATION

Everyone contributed **equally** to this project. The team collaborated through regular meetings, progress monitoring, and organized task management, ensuring that each part of the project was completed on schedule and that all elements were effectively integrated.